

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Streamlined Bakery Management: A Dedicated System for Cake Customization and Sales Monitoring

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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Data Informatics Track

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

Today, many businesses rely on technology to make their daily operations easier. However, many local businesses, such as Mr's Braves Cake owned by Mrs. Shamelyn Tapang, still use manual methods. Based on a recent interview, the shop has been relying on manual inventory and manual recording of transactions since 2020. Currently, customers can only inquire and place orders through the shop's Facebook page. Because custom cakes require specific details like tier sizes, edible paper, and fondant, relying on chat messages often leads to miscommunication. Furthermore, since the owner manages the business alone without extra manpower, manual scheduling and computing prices have become very difficult. This study proposes a Cake Shop Management System to help Mr's Braves Cake organize their custom orders through an add to cart and checkout process, process secure online payments, manage walk-in sales via a Point of Sale module, issue automated digital receipts, gather customer reviews, and track their daily ingredients.

1.2 Statement of the Problem

Currently, the manual process of managing Mr's Braves Cake causes several issues. Specifically, this study aims to solve the following problems:

1. Relying on Facebook messages for complex cake customizations like tier sizes and add-ons often leads to miscommunication and wrong pricing between the owner and the customer.
2. Manual recording of transactions and scheduling takes too much time, especially since the shop lacks additional manpower.
3. Tracking weekly sales and expenses manually is a slow process and is prone to calculation errors.
4. Customers have a hard time visualizing their preferences and knowing the estimated price range before placing an order.
5. The business lacks a unified checkout system to process full online payments and in-store Point of Sale transactions, issue automated receipts, and collect customer reviews.

1.3 Objectives of the Study

1.3.1 General Objective

The overall goal of the project is to design and develop a Cake Shop Management System for

Mr's Braves Cake that handles interactive custom orders, processes online payments and Point of Sale transactions, monitors inventory, and generates weekly sales and expense reports.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. To build an interactive order module where customers can input desired sizes, toggle add-ons like fondant or icing, upload inspiration pictures, and add items to a digital cart for checkout.
2. To integrate a secure online payment gateway that processes full payments instantly and automatically emails digital receipts to the customers.
3. To develop an in-store Point of Sale module allowing the administrator to input walk-in orders directly into the centralized system.
4. To develop an admin dashboard for the owner to track weekly sales, monitor expenses, easily add sample cakes or monthly bundles, and view customer reviews.
5. To create an inventory tracker that replaces the shop's manual 2020 recording system.

1.4 Scope and Delimitation

The system will cover an interactive customization tracker, an add to cart and checkout process, secure online payment processing, an automated email receipt generator, an in-store Point of Sale module, a customer review section, inventory monitoring, and weekly sales reporting. The system will only have two types of users: Mrs. Shamelyn Tapang as the sole Administrator, and the customers as the Users. What the system will not cover includes third-party delivery integrations like Grab, as the shop previously encountered tax-related issues with these platforms.

1.5 Significance of the Study

This project will be helpful to the following beneficiaries:

- **Administrator:** It gives Mrs. Shamelyn Tapang an easy way to view weekly sales, monitor expenses, and manage cake prices in one place without manually computing everything or relying on chat history.
- **Users:** It provides customers with a clear and interactive platform to select their preferences like dummy layers or tier sizes, upload inspiration pictures, check out securely using online payments, receive automated receipts, leave reviews, and track their orders without needing long Facebook conversations.
- **Researchers:** It serves as practical experience for BSIT students in building Systems, designing user interfaces, and managing databases.
- **Future Developers:** It can be used as a reference guide for future IT students studying small business systems and bakery e-commerce

1.6 Definition of Terms

- **Add-ons:** Extra cake decorations requested by the customer, such as edible paper or fondant, which affect the final price.
- **Administrator:** Refers to the business owner who manages the inventory, approves customer orders, and monitors the sales.
- **Customization Tracking:** The process of recording the specific design, flavor, tier size, and inspiration picture requested by the customer.
- **User:** Refers to the customers who will access the system to customize and place their cake orders.
- **Add to Cart:** A feature allowing users to temporarily save their customized cake before proceeding to checkout.
- **Checkout:** The final step in the online ordering process where the customer confirms details.
- **Online Payment:** A digital transaction method allowing customers to pay for their cakes securely in full.
- **Point of Sale:** An interface used by the administrator to punch in orders for physical walk-in customers.
- **Receipt:** An automated digital document sent to the customer's email after a successful transaction.
- **Review:** A feature allowing customers to leave feedback and ratings on their completed orders.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

The Digital Revolution in the Bakery Sector

A recent study published in the *MDPI Journal of Foods* emphasizes that transitioning from manual to digital operations allows bakeries to optimize inventory and reduce waste (Smith & Johnson, 2025). The researchers noted that digitalization leads to optimized inventory management, data-driven decision-making, and increased profitability, replacing the guesswork of manual tracking.

- **Relevance:** This study directly supports the need for Mrs. Shamelyn Tapang to upgrade her manual 2020 recording system. By applying a digital inventory tracker, Mr's Braves Cake can avoid wasted ingredients and compute expenses accurately.

Web-Based Cake Ordering Systems

A study by Williams et al. (2025) published in the *Journal of Innovation of Vocational Technology Education* focused on designing a web-based cake ordering platform. The researchers identified that most existing e-commerce platforms are too generic and fail to address the unique needs of a bakery, such as real-time product customization and integrated inventory.

- **Relevance:** This study validates the technical approach of the proposed system. It proves that custom cake shops require a dedicated platform where users can easily customize their orders, rather than using a standard retail point-of-sale system.

2.2 Local Studies

Strategic Management in the Philippine Cake Industry

A local study by Cruz and Santos (2025) from the *Cognizance Journal* examined the behavior of local cake consumers. The findings revealed that while local customers highly value customized offerings (like specific themes and tier sizes), local cake businesses often lack a structured, data-driven platform to manage these personalized requests efficiently.

- **Relevance:** The proposed system will feature an interactive order module that perfectly handles the structured customization that Philippine consumers look for, bridging the gap identified by this study.

The Shift of Philippine SMEs from Facebook to Dedicated Platforms

A study by Mendoza (2024) for the *IEOM Society* investigated how local small businesses heavily rely on Facebook for transactions. The study concluded that as a business grows, relying entirely on a social media chat interface creates friction, delayed replies, and

disorganized customer records.

- **Relevance:** This perfectly describes the current situation of Mr's Braves Cake. Because Mrs. Tapang manages the shop alone, relying on Facebook chats to finalize complex cake designs and prices is no longer sustainable.

2.3 Related Literature

Systems Analysis and the Shift to Automation

In the book *Modern Systems Analysis and Design*, Valacich and George (2020) explain that transitioning a business from a manual, paper-based recording system to an automated information system is critical for reducing human error and increasing operational speed. The authors emphasize that manual systems become severe bottlenecks as a business scales. For Mr's Braves Cake, replacing the manual 2020 recording notebook with an automated web system directly aligns with this principle, allowing the sole administrator to handle more orders without being overwhelmed by administrative tasks.

Cloud Database Management

According to the textbook *Database Systems: Design, Implementation, and Management* by Coronel and Morris (2019), modern businesses are increasingly relying on cloud-based database architectures to ensure data availability and real-time processing. The authors note that cloud databases allow for seamless updates across different platforms without requiring the business owner to maintain expensive physical servers. This literature justifies the use of Google Firestore for the cake shop system, as it provides the necessary real-time synchronization to instantly deduct ingredient stocks the moment a customer submits an order.

E-Commerce for Small Enterprises

In *E-Commerce: Business, Technology, Society*, Laudon and Traver (2021) discuss how small businesses must eventually transition from informal social media transactions to dedicated web platforms. The book highlights that while social media messaging is good for initial marketing, it lacks the structured data collection needed for complex orders. This perfectly describes the gap the proposed system fills. By moving customers away from Facebook chats and into a structured web interface, the cake shop can capture specific customization details accurately and automatically generate invoices.

Inventory Management for Perishable Goods

Muller (2019), in the book *Essentials of Inventory Management*, stresses that businesses dealing with perishable materials, such as bakery ingredients, must have precise and real-time inventory tracking. The author explains that relying on end-of-day or end-of-week manual counts often leads to stockouts or ingredient spoilage. Integrating an automated inventory deduction feature into the ordering process ensures that Mr's Braves Cake maintains exact, up-to-the-minute stock levels, preventing wasted ingredients and lost sales.

2.4 Synthesis of the Reviewed Literature

The gathered academic studies and published literature clearly show that manual inventory and chat-based ordering are only effective in the very early stages of a business (Mendoza, 2024). Both foreign and local researchers, alongside established systems analysis principles, agree that as a bakery grows, the lack of an automated system leads to miscommunication, wasted ingredients, and inefficient sales tracking (Smith & Johnson, 2025; Valacich & George, 2020). Furthermore, standard inventory management and e-commerce literature emphasizes the necessity of real-time cloud databases to prevent stockouts of perishable goods (Muller, 2019; Coronel & Morris, 2019). While there are many general point-of-sale systems available, most do not support the complex needs of a custom cake shop. The proposed system addresses this gap by allowing the customer to interactively build their cake online and automatically calculate the total price, effectively replacing Mr's Braves Cake's outdated manual setup.

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CHAPTER III

METHODOLOGY

3.1 Research Design

The study will use a developmental research design, focusing on the design, development, and evaluation of the proposed Cake Shop Management System.

3.2 System Development Methodology

The Agile Development Model will be used, consisting of the following phases:

1. Requirements Analysis
2. System Design
3. Development
4. Testing
5. Deployment (Prototype)

3.3 System Architecture

The system will follow a client-server architecture with cloud integration. Users will interact through a web interface, while the backend server will handle data processing, online payments, Point of Sale entries, price calculations, and receipt generation. Data will be instantly synced and managed through a real-time cloud database.

3.4 Tools and Technologies Used

Frontend: HTML, CSS, JavaScript, Bootstrap

Backend: Python

Database: Google Firestore / Firebase

Development Tools: Visual Studio Code

3.5 Data Gathering Techniques

Interviews: Conducted with Mrs. Shamelyn Tapang to determine exact pricing models, add-on costs (e.g., fondant, icing), and current manual challenges. Observation: Reviewing how Mr's Braves Cake currently handles Facebook inquiries and manual inventory notebooks.

3.6 Testing Procedures

Unit Testing: Testing individual features like the online checkout, Point of Sale, and customer review modules.

System Testing: Testing the complete flow from order placement to automated receipt emailing.

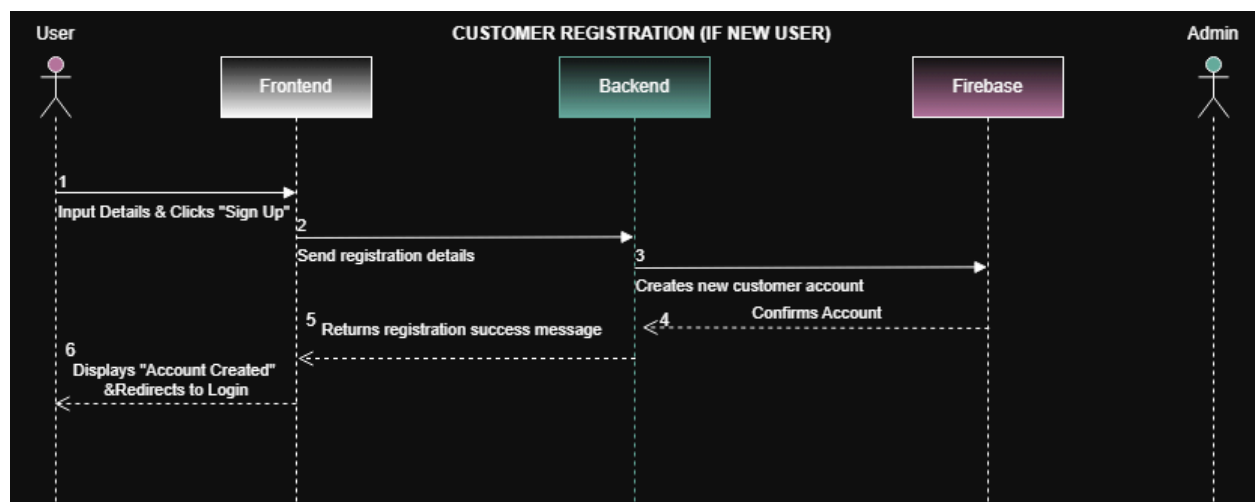
User Acceptance Testing (UAT): Allowing the admin and sample customers to use the system to ensure it meets real-world needs.

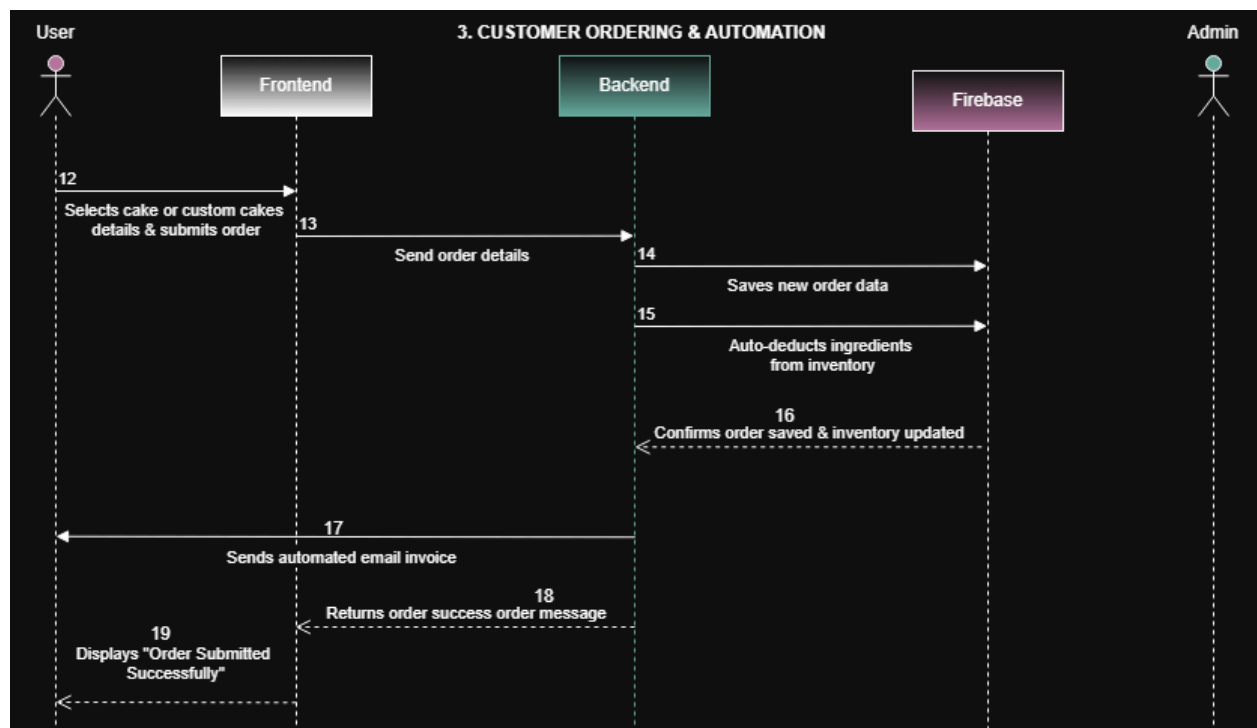
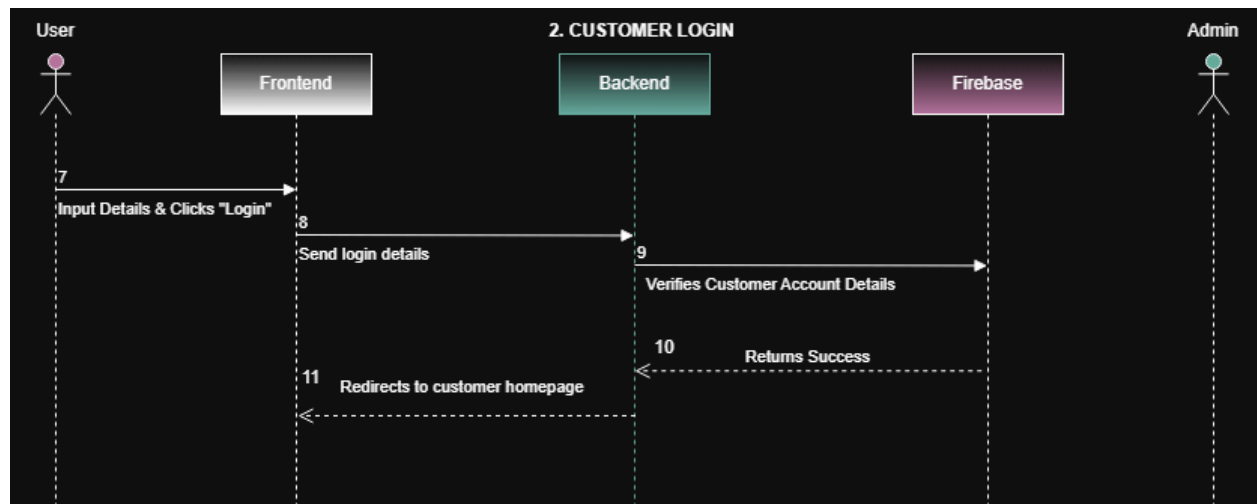
3.7 Evaluation Criteria

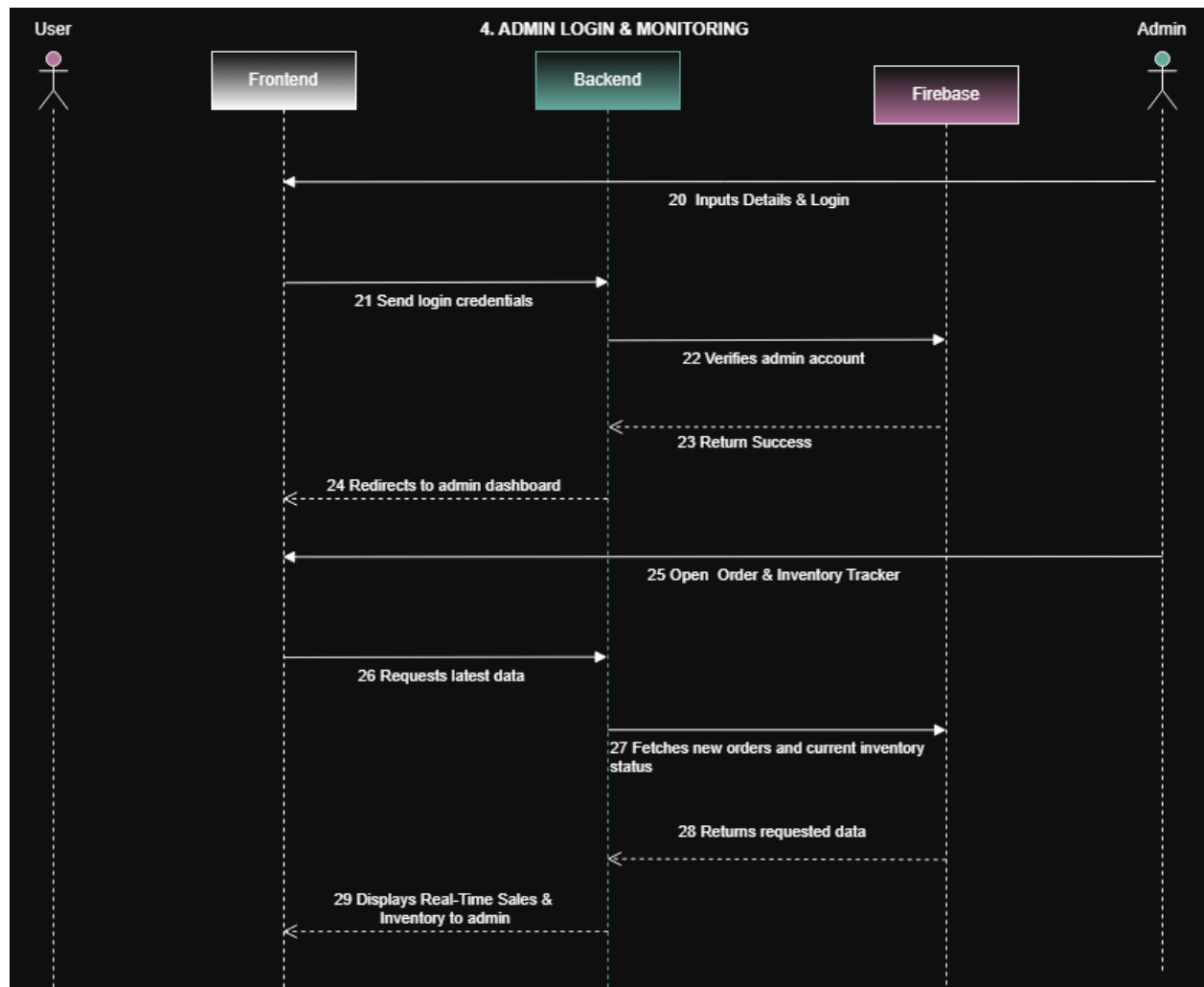
The system will be evaluated based on the following criteria: Functionality Usability Security (Protection of online payments and data) Efficiency

SUPPORTING DIAGRAMS

Because the project utilizes a NoSQL database architecture through Google Firestore, traditional relational entity diagrams are not applicable. Instead, the system's process and data flow are mapped out entirely using a Sequence Diagram. This diagram illustrates the exact interactions between the customer, the web interface, the Python backend, the cloud database, and the administrator. It details the step-by-step execution of the interactive selection of cake add-ons, the checkout and online payment process, the generation of automated email receipts, and the instant inventory deduction required for accurate real-time monitoring.







End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in System development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.